SOC Lab Report: PowerShell Execution Detection using Splunk

Lokeshwar V

1 Abstract

This report demonstrates detection of suspicious PowerShell execution in a SOC environment using Splunk. Attackers commonly use PowerShell for post-exploitation activities. This scenario simulates encoded command execution and analyzes logs to detect malicious behavior.

2 Objective

To detect suspicious PowerShell execution using process creation logs in Splunk.

3 Lab Setup

- Attacker Machine: Kali Linux
- Victim Machine: Windows 11
- SIEM Tool: Splunk (Ubuntu Server)
- Network: Internal Virtual Lab

4 MITRE ATT&CK Mapping

- Tactic: Execution
- Technique: PowerShell (T1059.001)

5 Attack Simulation

A suspicious PowerShell command was executed using an encoded payload.

5.1 Command Used

```
powershell -EncodedCommand VwByAGkAdABlAC0ATwB1AHQAcAB1AHQAIAAiAE0AYQBsAGkA
```

5.2 Explanation

The command executes a Base64 encoded payload. Attackers use encoded commands to evade detection.

5.3 Screenshot

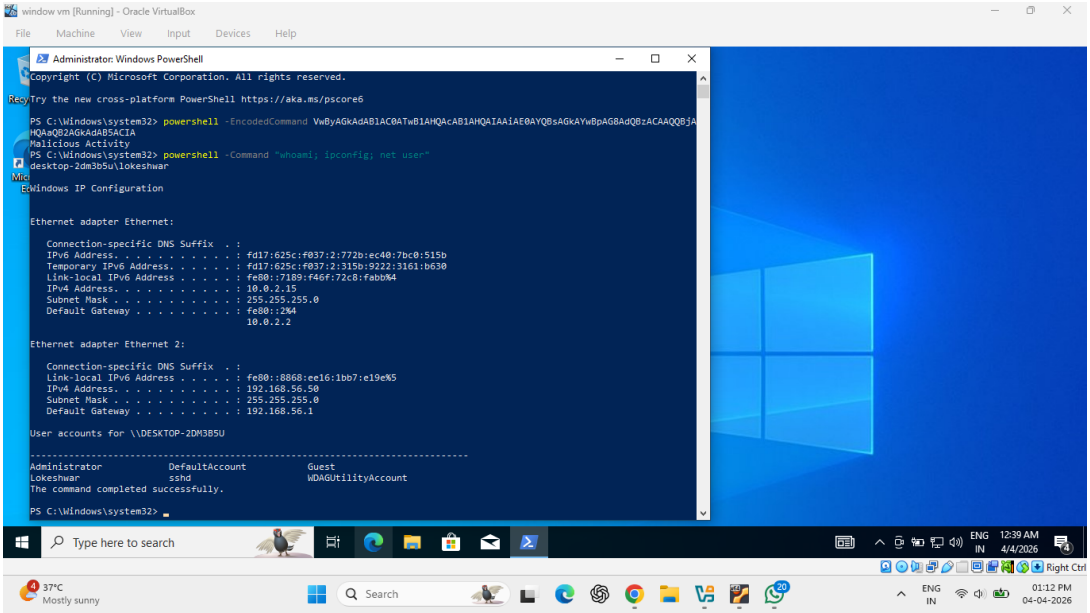

Figure 1: Execution of Encoded PowerShell Command

6 Detection in Splunk

PowerShell execution is logged using EventCode 4688.

6.1 Detection Query

```
index=windows EventCode=4688
| search Command_Line="*EncodedCommand*"
| table _time, user, New_Process_Name, Command_Line
```

6.2 Explanation

The query identifies PowerShell processes executing encoded commands, which are commonly used in attacks.

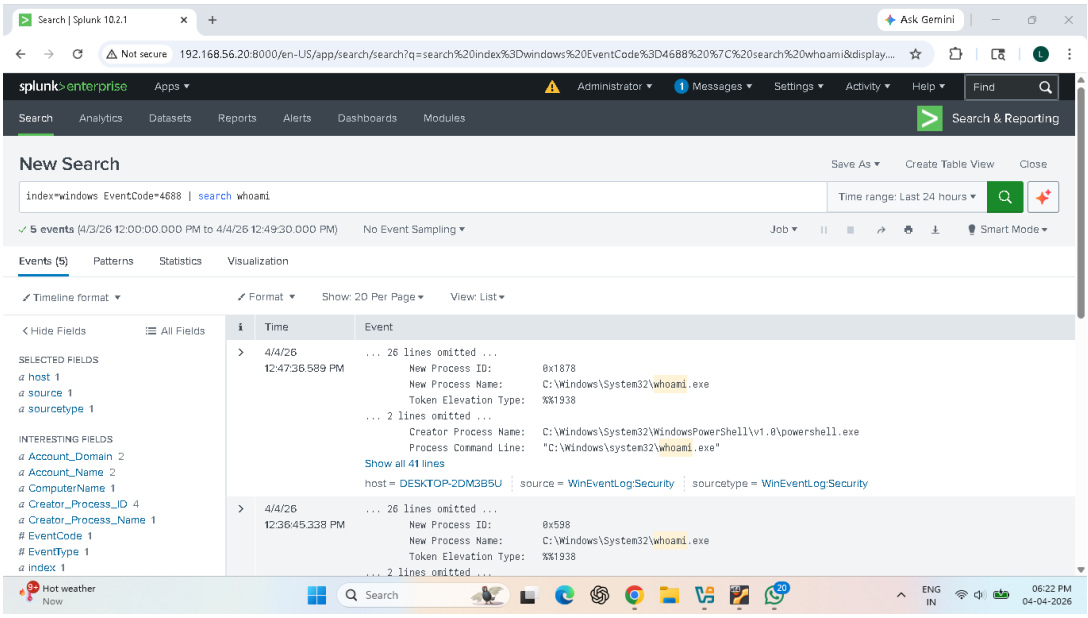

Figure 2: Detection of Encoded PowerShell Execution

6.3 Screenshot

6.4 Screenshot

7 Investigation

Further investigation was performed to analyze the executed commands.

7.1 Timeline Query

```
index=windows user="TARGET_USER"
| sort _time
| table _time, EventCode, Command_Line
```

7.2 Findings

- PowerShell process executed with encoded command
- Command obfuscation detected
- Possible malicious intent

7.3 Screenshot

8 Response Actions

Immediate steps were taken to mitigate the threat.

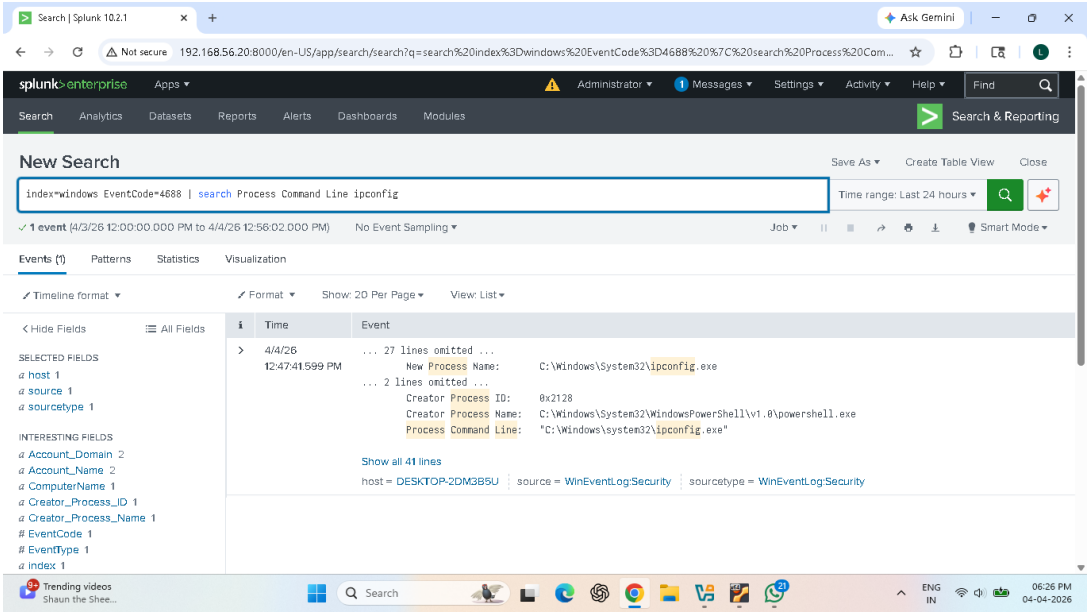

Figure 3: Detection of Encoded PowerShell Execution

- Terminated PowerShell process
- Isolated the system
- Reset user credentials

8.1 Commands Used

```
taskkill /IM powershell.exe /F
netsh interface set interface "Ethernet" admin=disable
net user <username> NewPassword@123
```

8.2 Explanation

The malicious process was terminated to stop execution. The system was isolated to prevent further communication, and user credentials were reset.

8.3 Screenshot

9 Incident Reporting

9.1 Incident Summary

Suspicious PowerShell execution using encoded commands was detected, indicating possible post-exploitation activity.

9.2 Impact

Potential execution of malicious scripts and system compromise.

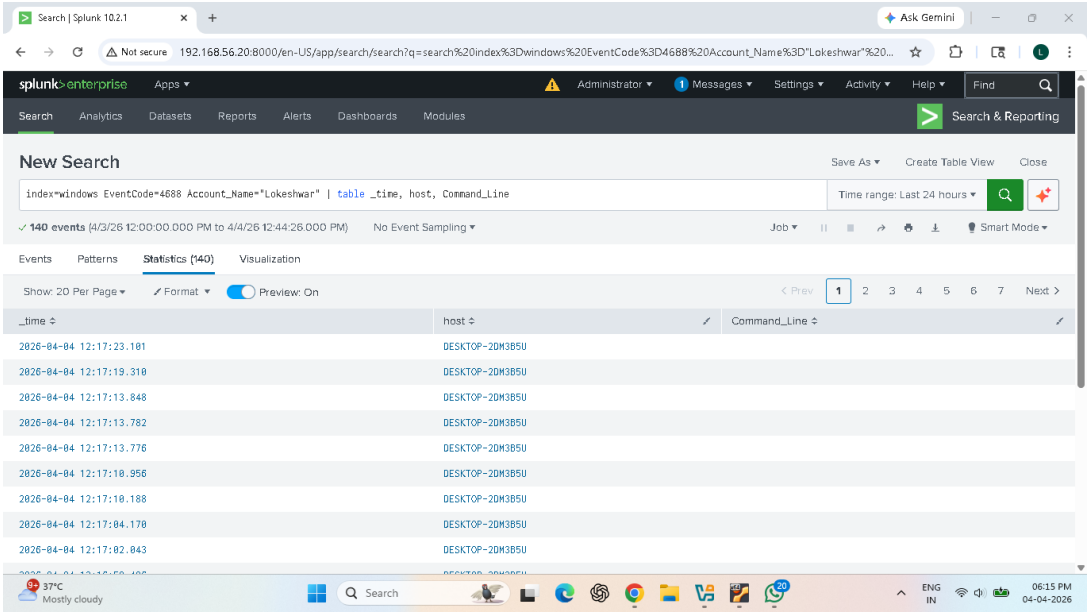

Figure 4: Timeline showing PowerShell activity

9.3 Conclusion

The attack was successfully detected and mitigated using Splunk SIEM.

10 Conclusion

This scenario highlights how attackers use PowerShell for malicious activities and how such behavior can be detected and handled effectively.

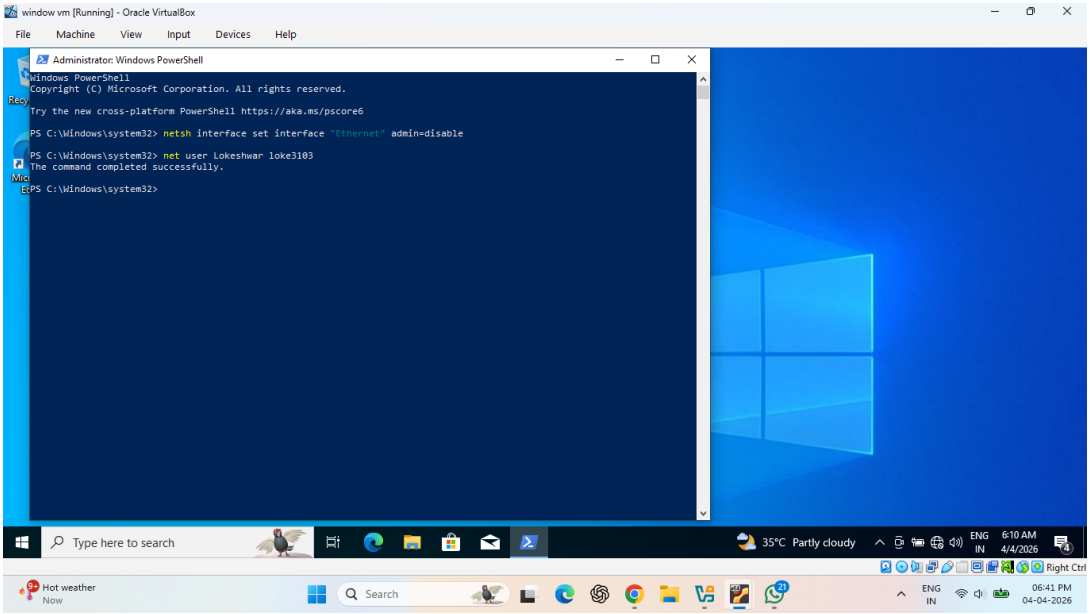

Figure 5: Response Actions - Process Termination and Isolation